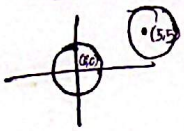
Two - Dimentional Transformation :

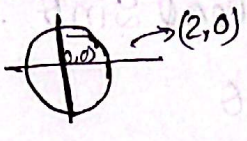
Geometric Transformation

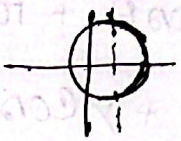
1. Object moves
2. Background fixed



Coordinate Transformation

1. Object fixed
2. Background moves.



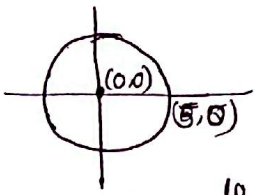


Types of Operations: (both are for Geometric and coordinate)

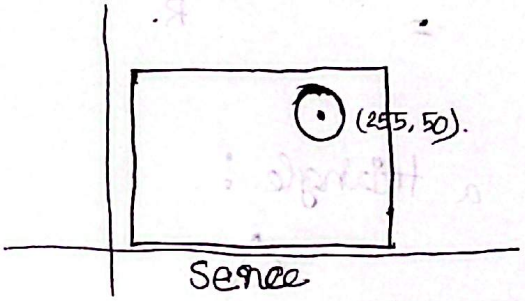
1. Translation.
2. Rotation.
3. Scaling
4. Mirroring.

Local Coordinates : Object's own space coordinates.

World Coordinates: Scene's coordinate where subjects are placed.

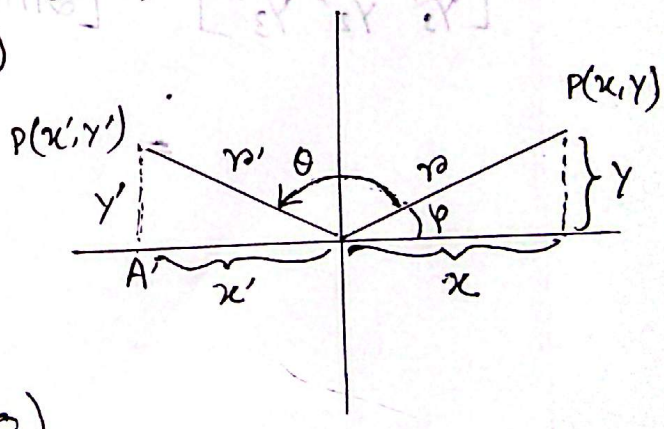


local cordinate.



word coodinate.

Rotation : Pivot (কার চারপাশে ঘুরাবো).
Angle (কত কোনে)



$\cos\phi = \frac{x}{r}$; $\therefore x = r\cos\phi$

$\sin\phi = \frac{y}{r}$; $\therefore y = r\sin\phi$

After rotation,

$x' = r\cos(\phi+\theta)$

$y' = r\sin(\phi+\theta)$

i $\Rightarrow x' = r\{\cos\phi\cos\theta - \sin\phi\sin\theta\}$.

$= r\cos\phi\cos\theta - r\sin\phi\sin\theta$.

$x' = x\cos\theta - y\sin\theta$.

ii $\Rightarrow y' = r\{\sin\phi\cos\theta + \cos\phi\sin\theta\}$.

$= r\sin\phi\cos\theta + r\cos\phi\sin\theta$.

$= x\sin\theta + y\cos\theta$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

A = R . B.

A = After Rotation
R = Rotation.
B = Before.

for a triangle :

△

$A = R \cdot B$.

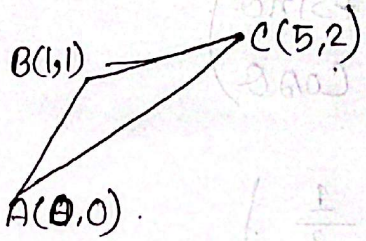
$$\begin{bmatrix} x_1' & x_2' & x_3' \\ y_3' & y_2' & y_3' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix}.$$

$=$

1. Find the Rotation matrix if an object is rotated by 30° when the pivot is origin.

2. What are the new coordinates of point P(2,-4) after rotation ?.

3. Perform 45° rotation around the origin of the triangle



Solve:

1. $R\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

1. Given, $\theta = 30°$.

$$R\theta = \begin{pmatrix} \cos 30 & -\sin 30 \\ \sin 30 & \cos 30 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$$= \begin{vmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{vmatrix}$$

$$= \left\{ \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \left(-\frac{1}{2}\right) \right\} = \frac{3}{4} + \frac{1}{4}$$

$$= 1.$$

Find the rotation matrix if the object is rotated by 30° clockwise around the origin.

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

$$= \begin{pmatrix} \cos(-\theta) & +\sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}.$$

$$= \left(\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}\right) - \left(-\frac{1}{2} \times \frac{1}{2}\right)$$

$$= \frac{3}{4} - \left(-\frac{1}{4}\right)$$

$$= \frac{3}{4} + \frac{1}{4} = 1.$$

2. Solve: $R_\theta = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \times \begin{pmatrix} 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 2+\sqrt{3} \\ 1-2\sqrt{3} \end{pmatrix}$

$$= \begin{bmatrix} \frac{\sqrt{3}}{2} \times 2 + \left(-\frac{1}{2} \times -4\right) \\ \frac{1}{2} \times 2 + \left(\frac{\sqrt{3}}{2} \times -4\right) \end{bmatrix} = \begin{bmatrix} \sqrt{3} & 2 \\ 1 & 2\sqrt{3} \end{bmatrix}$$

3. $R_{45} = \begin{pmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{pmatrix} \begin{pmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}_{2\times 2} \begin{pmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \end{pmatrix}_{2\times 3}$

$= \frac{1}{2}\times 0 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times 5$

$= \begin{pmatrix} \frac{1}{\sqrt{2}}\times 0 + -\frac{1}{\sqrt{2}}\times 0 & \frac{1}{\sqrt{2}}\times 1 + -\frac{1}{\sqrt{2}}\times 1 & \frac{1}{\sqrt{2}}\times 5 + -\frac{1}{\sqrt{2}}\times 2 \\ \frac{1}{\sqrt{2}}\times 0 + \frac{1}{\sqrt{2}}\times 0 & \frac{1}{\sqrt{2}}\times 1 + \frac{1}{\sqrt{2}}\times 1 & \frac{1}{\sqrt{2}}\times 5 + \frac{1}{\sqrt{2}}\times 2 \end{pmatrix}$

$= \begin{pmatrix} 0 & 0 & \frac{3}{\sqrt{2}} \\ 0 & \sqrt{2} & \frac{7}{\sqrt{2}} \end{pmatrix}$

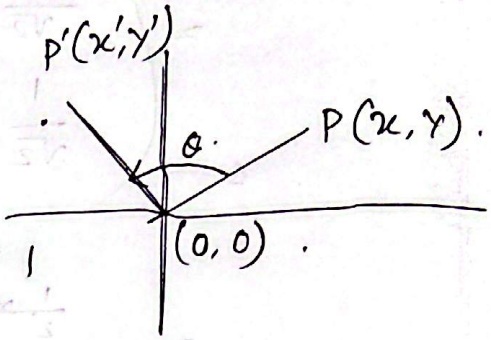
2D Transformation :

Rotation around origin :

If a point $p(x,y)$ is rotated anticlockwise θ° by origin, then new coordinates of P,

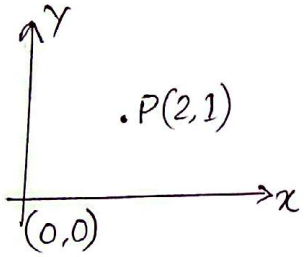
$$P' = R_\theta \cdot P$$

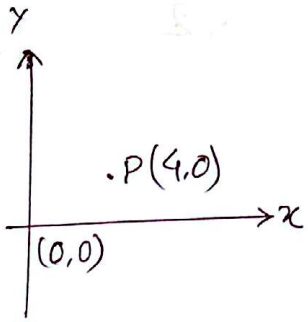
$$\begin{pmatrix} x' \\ y' \end{pmatrix}_{2\times 1} = \underbrace{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}_{\text{Rotation matrix}}{}_{2\times 2} \begin{pmatrix} x \\ y \end{pmatrix}_{2\times 1}$$

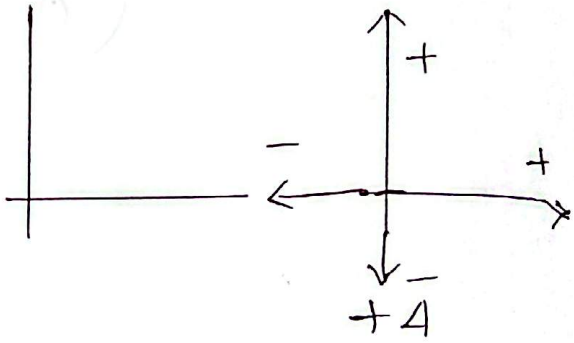


$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}_{3\times 1} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}_{3\times 3} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}_{3\times 1}$$

Translation : (Add)







$P \rightarrow \underline{tx} = +2$

$\underline{ty} = -1$

$x' = 4(x+tx)$

$Y' = 0(Y+ty)$

Rotation :

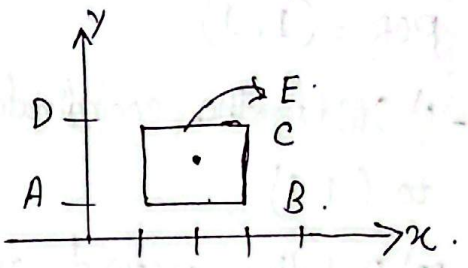
$P' = tr \cdot P$

↳ Transformation matrix.

→ transformation matrix.

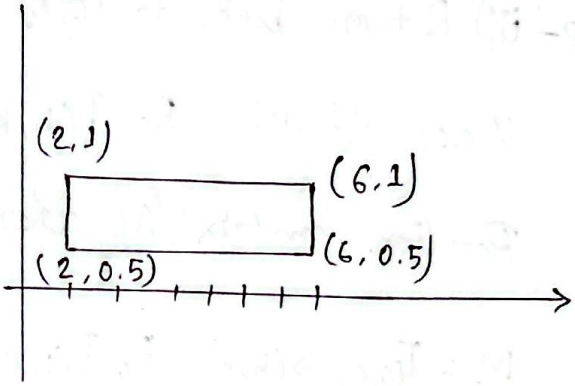
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x+0+tx \\ 0+y+ty \\ 0+0+1 \end{pmatrix} = \begin{pmatrix} x+tx \\ y+ty \\ 1 \end{pmatrix}.$$

Scaling around origin: Multiply



$O'(0,0) \rightarrow Sx = 2$.
$Sy = 1/2$.

A(1, 1) $Sx = 2$
B(3, 1) $Sy = 1/2$.
C(3, 2) A' (2, .5).
D(1, 2) B' (6, .5)
C' (6, 1)
$Sy = 1/2$ D' (2, 1).
$Sx = 2$



$x' = Sx \cdot x$; $y' = Sy \cdot y$.

$P' = S \cdot P$.

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} Sx.x + 0.y + 0.1 \\ 0.x + y.Sy + 0.1 \\ 0.x + 0.y + 1.1 \end{pmatrix}$$

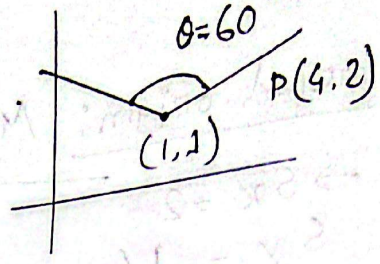
$$= \begin{pmatrix} Sx.x \\ Sy.y \\ 0 \end{pmatrix}$$

4.3 book.

R_0 = origin POR (0,0).

Random point :.

POR = (1, 1).



Step- i) Shift the coordinate system to (1, 1)

Step- ii) Rotation around new origin.

Step- iii) Return back to (0, 0) coordinate system.

$T(0,0) \rightarrow (h,K)$ R_0 $T(h,K) \rightarrow (0,0)$

$(1,1) \rightarrow (0,0)$
$tx = -1$
$ty = -1$

Rotation matrix M when POR is (h, K)

$$M = T_{(0,0)\rightarrow(h,K)} \; R_0 \; T_{(h,K)\rightarrow(0,0)}$$

$$= \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & K \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -h \\ 0 & 1 & -K \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & K \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta & -h\cos\theta + K\sin\theta \\ \sin\theta & \cos\theta & -h\sin\theta - K\cos\theta \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta & -\sin\theta & -h\cos\theta + K\sin\theta + h \\ \sin\theta & \cos\theta & -h\sin\theta - K\cos\theta + h \\ 0 & 0 & 1 \end{pmatrix}$$

= Final Formula.

4.5/ A(0, 0), B(1, 1) C(5, 2)

45° anticlockwise.

i) around (0, 0)

ii) around (-1, -1)

~~iii) Rotation Matrix = Cos45° - Sin45° (1)+(-1)~~

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & -h\cos\theta + k\sin\theta + h \\ \sin\theta & \cos\theta & -h\sin\theta - k\cos\theta + h \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 1/\sqrt{2} - 1/\sqrt{2} - 1 \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} + 1/\sqrt{2} - 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$P = R.P'$

$$= \begin{vmatrix} 1/\sqrt{2} & -1/\sqrt{2} & -1 \\ 1/\sqrt{2} & 1/\sqrt{2} & -1+\sqrt{2} \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \begin{pmatrix} 0+0-1 & 1/\sqrt{2} - 1/\sqrt{2} - 1 & 5/\sqrt{2} - \frac{2}{\sqrt{2}} - 1 \\ -1+\sqrt{2} & -1+2\sqrt{2} & \frac{-2+5\sqrt{5}}{2} \\ 1 & 1 & 1 \end{pmatrix}$$

4.6 Find the transformation of a point P(x,y) that

a) Scales a units in x direction.

b) Scales b units in y direction.

c) Simultaneously (a) and (b).

Find new transformat coordinates of p fore (a) (b) (c) separately.

$S = \begin{pmatrix} S_x & 0 \\ 0 & S_y \end{pmatrix}$

[transform matrix, ~~New~~ co ordinate.

[ordinate coordiant

$P'(x', y') = S \cdot P$.

$= \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$.

$= \begin{pmatrix} ax \\ y \end{pmatrix}$.

a) P(x, y)

$S_x = a$

$S_y = 1$.

b) $S = \begin{vmatrix} 1 & 0 \\ 0 & b \end{vmatrix}$.

$A = \begin{vmatrix} x \\ y \end{vmatrix}$.

$A' = SA$.

$= \begin{vmatrix} 1 & 0 \\ 0 & b \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} x \\ by \end{vmatrix}$.

c) $S = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix}$

$A = \begin{vmatrix} x \\ y \end{vmatrix}$

$A' = SA = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} ax \\ by \end{vmatrix}$

4.7. $S = \begin{pmatrix} S_x & 0 \\ 0 & S_y \end{pmatrix}$ ($S_x \rightarrow$ x direction scaling factor)

$$= \begin{pmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{pmatrix}. \qquad x' = S_x \cdot x. \quad y' = S_y \cdot y.$$

Scaling around the origin.

Scaling around a point $P(h,K)$.

$M = T_{(0,0)\rightarrow(h,K)} \quad S \cdot T_{(h,K)\rightarrow(0,0)}.$

$$= \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & K \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -h \\ 0 & 1 & -K \\ 0 & 0 & 1 \end{pmatrix}.$$

$$= \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & K \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_x & 0 & -S_x h \\ 0 & S_y & -S_y K \\ 0 & 0 & 1 \end{pmatrix}.$$

$$= \begin{pmatrix} S_x & 0 & -hS_x + h \\ 0 & S_y & -hS_y + K \\ 0 & 0 & 1 \end{pmatrix}.$$

A(1, 1) → C(5, 2). Twice magnify x → Scale
y → Scale.

Transformation matrix, $M = \begin{pmatrix} 2 & 0 & -5\times 2 + 5 \\ 0 & 2 & -2\times 2 + 2 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 2 & 0 & -5 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix}$.

$A' = M \cdot A$

$= \begin{pmatrix} 2 & 0 & -5 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$

A' Coordinate (-3, 0).